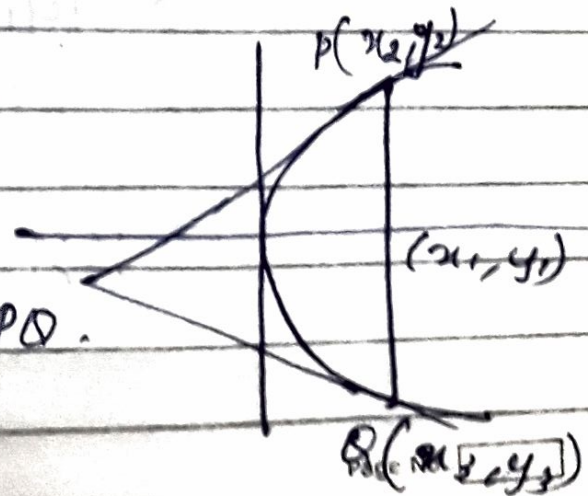


Chord with a given mid point.

Q Find the eqs of chord of: (a) parabola (b) ellipse (c) hyperbola having (x_1, y_1) as its mid point.

a) $y^2 = 4ax$

Let (x_1, y_1) be the mid point of chord ^{joining} $P(x_2, y_2)$ and $Q(x_3, y_3)$; PO .



$\therefore P$ and Q are points on parabola

$$\Rightarrow y_2^2 = 4ax_2 \quad \text{--- (1)}$$

$$\Rightarrow y_3^2 = 4ax_3 \quad \text{--- (2)}$$

$$(2)-(1) \Rightarrow y_3^2 - y_2^2 = 4a(x_3 - x_2)$$

$$\Rightarrow (y_3 + y_2)(y_3 - y_2) = 4a(x_3 - x_2)$$

$$\frac{y_3 - y_2}{x_3 - x_2} = \frac{4a}{y_3 + y_2} \quad \text{--- (3)}$$

$\therefore (x_1, y_1)$ is the mid point of PQ .

$$x_1 = \frac{x_2 + x_3}{2} \quad \& \quad y_1 = \frac{y_2 + y_3}{2} \quad \text{--- (4)}$$

Sub (4) in (3)

$$\Rightarrow \frac{y_3 - y_2}{x_3 - x_2} = \frac{4a}{2y_1} = \frac{2a}{y_1}$$

$\therefore PQ$ is a line passing through (x_1, y_1) with slope $\frac{2a}{y_1}$

$$\Rightarrow y - y_1 = \frac{2a}{y_1}(x - x_1)$$

$$\Rightarrow yy_1 - y_1^2 = 2a(x - x_1)$$

+ $(2ax_1)$ on both sides

$$\Rightarrow yy_1 - 2ax_1 - y_1^2 = 2ax - 2ax_1 - 2ax_1$$

$$\Rightarrow yy_1 - 2ax_1 - y_1^2 = 2ax - 4ax_1$$

$$\Rightarrow yy_1 - 2a(x + x_1) = y_1^2 - 4ax_1$$

$$\underline{\underline{T_1 = S_1}}$$

where $T_1 = yy_1 - 2a(x+x_1)$
 $S_1 = y_1^2 - 4ax_1$

b) Let (x_1, y_1) be the mid point of chord joining $P(x_2, y_2)$ and $Q(x_3, y_3)$

$\therefore P$ and Q are points on ellipse

$$\frac{x_2^2}{a^2} + \frac{y_2^2}{b^2} = 1 \quad (1)$$

$$\frac{x_3^2}{a^2} + \frac{y_3^2}{b^2} = 1 \quad (2)$$

$$(2) - (1) \Rightarrow \frac{1}{a^2}(x_3^2 - x_2^2) + \frac{1}{b^2}(y_3^2 - y_2^2) = 0$$

$$= \frac{x_3^2 - x_2^2}{a^2} = - \frac{(y_3^2 - y_2^2)}{b^2}$$

$$\Rightarrow \frac{y_3 - y_2}{x_3 - x_2} = - \frac{b^2}{a^2} \frac{x_3 + x_2}{y_3 + y_2} \quad (3)$$

$\therefore (x_1, y_1)$ is the mid point of PQ

$$x_1 = \frac{x_2 + x_3}{2} \quad y_1 = \frac{y_2 + y_3}{2} \quad (4)$$

Sub (4) in (3)

$$\Rightarrow \frac{y_3 - y_2}{x_3 - x_2} = - \frac{b^2}{a^2} \frac{2x_1}{2y_1}$$

$$\text{Slope is } - \frac{b^2}{a^2} \frac{x_1}{y_1}$$

$$\Rightarrow y - y_1 = - \frac{b^2}{a^2} \frac{x_1}{y_1} (x - x_1)$$

$$\Rightarrow yy_1 a^2 - y_1^2 a^2 = -b^2 x x_1 + b^2 x_1^2$$

$\div$ by $a^2 b^2$

$$\frac{yy_1}{b^2} - \frac{y_1^2}{b^2} = \frac{x_1^2}{a^2} - \frac{xx_1}{a^2}$$

$$\frac{xx_1}{a^2} + \frac{yy_1}{b^2} = \frac{x_1^2}{a^2} + \frac{y_1^2}{b^2}$$

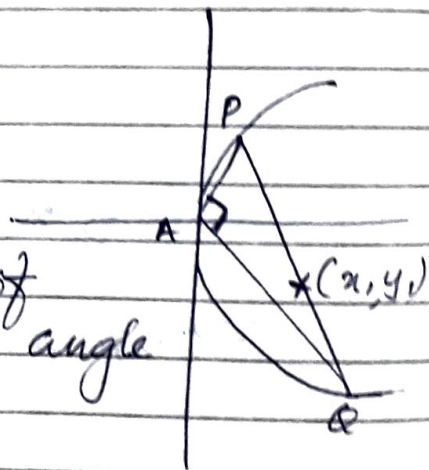
$$\underline{\underline{T_1 = S_1}}$$

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Q. P.T locus of mid points of chords of a parabola will subtend a right angle at vertex is another parabola of half the latus rectum of original parabola.

Let the given parabola be
 $y^2 = 4ax$ — (1)

Let (x_1, y_1) be the mid point of chord PQ which subtend right angle at vertex. $\angle$



$$T_1 = S_1$$

$$yy_1 - 2a(x+x_1) = y_1^2 - 4ax_1$$

$$yy_1 - 2ax = y_1^2 - 2ax_1$$

$$\frac{yy_1 - 2ax}{y_1^2 - 2ax_1} = 1 \quad \text{--- (2)}$$

$$y_1^2 - 2ax_1$$

The combined eqn of AP and AQ obtained by homogenising (1) and (2)

$$\Rightarrow y^2 = 4ax_1 \cdot 1$$

$$y^2 = 4ax_1 \left(\frac{yy_1 - 2ax_1}{y_1^2 - 2ax_1} \right)$$

$$y^2 y_1^2 - 2ay^2 x_1 = 4ax_1 yy_1 - 8a^2 x_1^2$$

$$8a^2 x^2 - 4axy y_1 + y^2 (y_1^2 - 2ax_1) = 0 \quad \text{--- (3)}$$

which is the combined eqn of AP and AQ

$$\therefore \angle PAQ = 90^\circ$$

from (3) $\text{Coeff.}(x^2) + \text{Coeff.}(y^2) = 0$

$$\Rightarrow 8a^2 + y_1^2 - 2ax_1 = 0$$

$$\Rightarrow y_1^2 = 2ax_1 - 8a^2$$

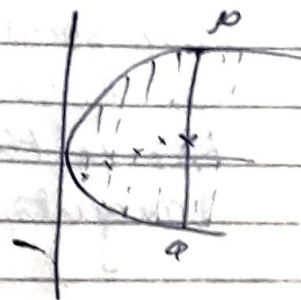
$$\Rightarrow y_1^2 = 2a(x_1 - 4a) \quad \text{--- (4)}$$

eqn (4) is a parabola with vertex $(4a, 0)$ and latus rectum $2a$.

Hence proved.

Q. Find locus of middle points of a system of parallel chords of the parabola $y^2 = 4ax$

Let the chords be $\parallel$ to the line $y = mx + c$ --- (1)



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Let (x_1, y_1) be the mid points of one of such chords PQ.

$\therefore$ Eqn of PQ is

$$T_1 = S_1$$

$$xy_1 - 2a(x+x_1) = y_1^2 - 4ax_1$$

$$xy_1 - 2ax = y_1^2 - 2ax_1$$

$$y = \frac{2ax}{y_1} + \frac{y_1^2 - 2ax_1}{y_1} \quad \text{--- (2)}$$

$\therefore$ (1) and (2) are $\parallel$ el, their slopes are equal

$$m = \frac{2a}{y_1}$$

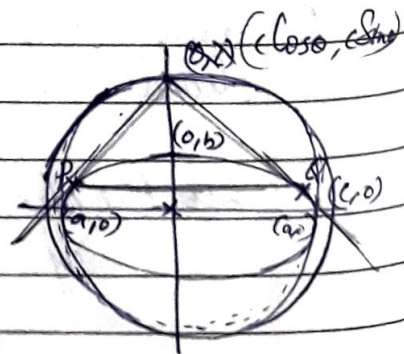
$$\Rightarrow y_1 = \frac{2a}{m}$$

$\therefore$ locus of (x_1, y_1) is $y = \frac{2a}{m}$ which is a

straight line $\parallel$ el to axis of parabola

Imp Q. Tangents are drawn to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ from points on the circle $x^2 + y^2 = c^2$. S.T the middle point of point of contact line on the curve $\left(\frac{x^2}{a^2} + \frac{y^2}{b^2}\right)^2 = \frac{x^2 + y^2}{c^2}$

any point on the circle $x^2 + y^2 = c^2$ is $(c \cos \theta, c \sin \theta)$. Then PQ is the chord of contact of tangents from $(c \cos \theta, c \sin \theta)$.



$$\therefore \frac{x x_1}{a^2} + \frac{y y_1}{b^2} = 1$$

$$\frac{x \cos \theta}{a^2} + \frac{y \sin \theta}{b^2} = 1 \quad \text{--- (1)}$$

Let (x_1, y_1) be the mid point of PQ. Then eqn of PQ is

$$T_1 = S_1$$

$$\frac{x x_1}{a^2} + \frac{y y_1}{b^2} = \frac{x_1^2}{a^2} + \frac{y_1^2}{b^2} \quad \text{--- (2)}$$

$\therefore$ (1) & (2) represent the same line.

$$\frac{\frac{c \cos \theta}{a^2}}{\frac{x_1}{a^2}} = \frac{\frac{c \sin \theta}{b^2}}{\frac{y_1}{b^2}} = \frac{1/\frac{x_1^2}{a^2} + \frac{y_1^2}{b^2}}{\frac{x_1^2}{a^2} + \frac{y_1^2}{b^2}}$$

$$\frac{c \cos \theta}{x_1} = \frac{c \sin \theta}{y_1} = \frac{x_1 a^2 b^2}{b^2 x_1^2 + a^2 y_1^2}$$

$$\cos \theta = \frac{x_1 a^2 b^2}{(b^2 x_1^2 + a^2 y_1^2)}$$

$$\sin \theta = \frac{y_1 a^2 b^2}{(b^2 x_1^2 + a^2 y_1^2)}$$

$$\cos \theta = \frac{x_1/c}{x_1^2/a^2 + y_1^2/b^2}$$

$$\sin \theta = \frac{y_1/c}{x_1^2/a^2 + y_1^2/b^2}$$

$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\left(\frac{y_1/c}{x_1^2/a^2 + y_1^2/b^2} \right)^2 + \left(\frac{x_1/c}{x_1^2/a^2 + y_1^2/b^2} \right)^2 = 1$$

$$\left(\frac{y_1}{c} \right)^2 + \left(\frac{x_1}{c} \right)^2 = \left(\frac{x_1^2}{a^2} + \frac{y_1^2}{b^2} \right)$$

$$\frac{x_1^2 + y_1^2}{c^2} = \left(\frac{x_1^2}{a^2} + \frac{y_1^2}{b^2} \right)$$

hence proved.